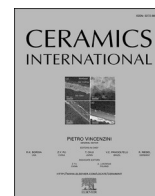




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Water corrosion mechanisms in ternary sodium borosilicate glass under varied pH conditions

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ARTICLE INFO

Handling Editor: Dr P. Vincenzini

Keywords:

Borosilicate glass
Leaching behavior
pH effect
Water corrosion mechanism

ABSTRACT

Investigating glass leaching behavior is essential to ensure the safety of high-level radioactive waste (HLW) vitrification during long-term deep geological disposal. This study elucidates water corrosion mechanisms in ternary sodium borosilicate glass by leaching across varied pH conditions (pH 3, 4, 5, 6.5, 9, 11, and 12). The pH value was adjusted with HCl and KOH solutions and the leaching experiments through the MCC-1 method at 90°C. Inductively coupled plasma optical emission spectrometer, Fourier-Transform infrared spectroscopy, X-ray photoelectron spectroscopy, and Time-of-flight secondary ion mass spectrometry were used to analyze the leachate and leached samples. The leaching behavior of silicon, boron, and sodium was meticulously assessed. Results indicate silicon's resistance to leaching under a wide range of pH levels, barring strong alkalinity where glass network integrity is compromised. Boron exhibited a pH-sensitive leaching pattern, the dominant mechanism is solid-liquid interdiffusion in acidic (pH 3 and 4), and neutral (pH 5, 6.5, and 9) conditions, while the main mechanism changed into network hydrolysis at alkaline conditions (pH 11 and 12). The leaching behavior of sodium was markedly affected by acidic conditions due to ion exchange. In strong alkaline conditions (pH 11 and 12), the glass network depolymerization increases the release of main elements in glass. This investigation sheds light on the complex interplay between glass composition and the environmental pH of water. It underscores the critical need to understand water corrosion mechanisms to ensure the long-term stability and safety of borosilicate glass used in HLW management.

1. Introduction

Borosilicate glass is universally recognized as the candidate material for immobilizing high-level radioactive waste (HLW), due to its durability and stability [1–3]. Despite these advantages, the incorporation of vitrification still within a comprehensive, multi-barrier containment system designed to mitigate the release of radioactive nuclides [4–8], concerns persist regarding the interaction of borosilicate glass containing nuclear waste with groundwater in the context of long-term geological disposal [9,10]. The integrity of borosilicate glass, particularly under the combined stresses of irradiation and leaching, is crucial for the safe disposal of HLW. Therefore, understanding the alteration behavior of borosilicate glasses upon water exposure is imperative for

assessing the long-term safety of HLW disposal strategies.

Over the past few decades, leaching experiments have been conducted under various experimental conditions (including the MCC-1 method [11–14], product consistency test (PCT) method [15,16], and vapor hydration test (VHT) method [17,18], among others) to investigate the long-term stability of nuclear waste glass. The reaction mechanisms between the nuclear waste glass and water have been extensively studied, leading to the development of pertinent models [19–24]. The leaching behavior of nuclear waste glass is influenced by numerous factors, including glass composition [25,26], temperature [27], pH of leaching solution [28], and the type of water (deionized or groundwater) [29,30]. Gin et al. [31] investigated the alteration kinetics of SON68 nuclear glass between pH 7 and 11.5. The results showed that a

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<https://doi.org/10.1016/j.ceramint.2025.01.352>

Received 2 October 2024; Received in revised form 25 December 2024; Accepted 19 January 2025

Available online 20 January 2025

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protective gel formed in all pH values but the stability of the gel was pH-dependent, it is more stable at pH < 10. Fournier et al. [32] studied the effects of pH (pH 1–10.7) on the stability of gel on International simple glass (ISG) glass. The finding reveals that the chemical duration decreases in the strong alkaline solution (pH > 9.5 at 90°C). Strachan [33] established glass dissolution as a function of pH, helping to understand mechanisms and guide future experiments. Pierce et al. [28] highlighted the effects of pH and temperature on dissolution rates of simulated aluminoborosilicate waste glasses. They found that matrix dissolution became the dominant dissolution mechanism for all glasses at the forward rate and pH values greater than 9. Vienna et al. [34] researched the impact of glass composition, pH, and temperature on glass forward dissolution rate. The works mentioned above elucidated aspects of the effect of pH on the formation of gel and the glass leaching behavior. However, most of these studies have been performed under alkaline conditions or in silicon-saturated solutions, with relatively fewer reports focusing on the leaching behavior in acidic solutions specifically during the initial leaching stages.

There have been many reports on the leaching behavior of different elements in the glass network. Gin et al. [35] revealed the fate of silicon in ISG glass during glass corrosion under alkaline conditions. Ojovan et al. [36] pointed out that ion exchange between alkali metal cations and protons from water is the principal mechanism if temperatures are low and pH is neutral. Farid et al. [37] addressed the evolution of cations speciation in the interfacial water species with the progress of the leaching process. Chave et al. [38] identified solid-state diffusion of both boron and alkali within glass and its amorphous alteration layer during nuclear glass residual alteration in solution. Zapol et al. [39] calculated reaction barriers for hydrolysis reactions on the glass surface using density functional theory (DFT), which indicates that reaction barriers in acidic conditions for the dissolution of boron-oxy-boron (B-O-B) and boron-oxy-silicon (B-O-Si) bridges are greatly reduced, compared with neutral and basic conditions. However, no comprehensive reports exist on the leaching behavior of the main elements (B, Si, and Na) in the glass during the initial leaching stages of borosilicate glass, particularly the leaching behaviors at different pH levels.

Elucidating the primary mechanisms governing glass-water interactions is critical to ensuring the efficacy and safety of immobilization strategies for HLW. Employing leachates of varying pH values is instrumental, as it elucidates the environmental impact on glass leaching dynamics. Nuclear waste glass exhibits a complex composition, often necessitating the simplification of its glass components for research. ISG is frequently utilized globally as a surrogate for nuclear waste glass. To enhance the understanding of the leaching behavior, ternary sodium borosilicate glass further streamlines the glass components presenting the distinct roles of glass network formers, intermediates, and modifiers in the leaching process. This paper investigates a ternary borosilicate glass to understand the effects of various pH environments on its leaching behavior, specifically focusing on the elemental constituents of Si, B, and Na. Utilizing leaching solutions formulated with analytically pure HCl and electronic grade KOH, the study aims to elucidate the mechanisms of water corrosion under acidic and alkaline conditions. This investigation provides critical insights into optimizing the long-term stability and environmental safety of borosilicate glasses used in the containment of HLW.

2. Materials and experiment

2.1. Sample preparation

Borosilicate glass was prepared via melting-annealing method [40, 41]. The dimension of the glasses was $10 \times 10 \times 1 \text{ mm}^3$ and underwent polishing on both sides to facilitate subsequent leaching experiments. Table 1 shows the chemical compositions and the density of the designed borosilicate glass.

Table 1

The chemical compositions and density of ternary borosilicate glass.

mol%	SiO ₂	B ₂ O ₃	Na ₂ O	Density (g/cm ³)
NBS5	67.26	18.60	14.14	2.44

2.2. Leaching agent preparation

In this study, deionized water served as the neutral leaching agent and primary water for adjusting the initial pH of the leaching agent, with its conductivity maintained at a maximum of 0.056 $\mu\text{S}/\text{cm}$. Before the leaching experiment, the deionized water was exposed to an atmospheric environment for over 30 minutes to ensure CO₂ saturation, the pH value is approximately 6.5. After preparation, the leaching agent underwent analysis via Inductively Coupled Plasma Optical Emission Spectrometry (ICP-OES), confirming that the background levels of common glass elements such as Si, B, and Na were well below the detection threshold, thus negating any concerns regarding background contamination.

For the acidic environment leach solution, a 37% HCl solution (Sigma-Aldrich, USA) was diluted with the CO₂ saturation deionized water. This preparation was characterized by a sulfate (SO₄²⁻) trace anion concentration of less than 1 ppm and a trace cation concentration of heavy metal elements also below 1 ppm, ensuring minimal potential interference with experimental outcomes. Conversely, the alkaline environment leach solution was generated by dissolving semiconductor-grade KOH pellets (Sigma-Aldrich, USA) in prepared deionized water. The KOH employed was of high purity, verified through Trace Metals Analysis to be 99.99 % pure, with sodium (Na) impurities in the raw material not exceeding 500 ppb.

The experimental acidic leaching solutions were meticulously prepared using hydrochloric acid (A.R. HCl) mentioned above. The initial solution, aimed to achieve a pH value of 2, was formulated by calculating and mixing the required quantity of HCl with deionized water at room temperature. Subsequently, employing a methodical step-by-step dilution technique, 100 ml of this highly concentrated acidic solution was transferred into a 1 L volumetric flask for the initial dilution phase. This process was repeated several times to systematically produce leaching agents with the desired experimental pH levels of 3, 4, and 5 at room temperature, respectively. Throughout the preparation phase, the deionized water employed was pre-saturated with CO₂, ensuring its utility as a neutral leaching solvent conducive to the experimental conditions. An analogous methodology was employed to formulate alkaline leaching solutions with pH values of 9, 11, and 12, respectively.

2.3. Leaching experiment

The sample was suspended in 36 ml of deionized water using a nylon thread in a polytetrafluoroethylene (PTFE) reactor (sealed within a stainless-steel container) at 90°C. The ratio of water volume to the sample surface area was 15 cm. To obtain the dissolution rate during the initial alteration stage, the samples were taken at 2, 3, 4, 6, 8, 12, 18, and 24 h, respectively.

The borosilicate glasses were synthesized under rigorously identical conditions to ensure experimental consistency and reliability. To minimize the uncertainties typically introduced by repeated sampling from a single source, each data point within the leaching experiment was derived from a distinct container. Consequently, nine individual pieces of glass samples were utilized to obtain a comprehensive set of data. This methodological choice significantly enhances the robustness of our findings by reducing potential variations and ensuring that each data point reflects a true measure of the leaching behavior under study.

2.4. Characterization methods

The elemental concentration in leachate was characterized by ICP-

OES (Plasma Quant PQ 9000 Elite, Analytik Jena, Jena, Germany). The leaching rate was calculated as following equations:

$$NL_i = \frac{C_i}{f_i \times \left(\frac{SA}{V}\right)} \quad (1)$$

where NL_i is the normalized mass loss of the element i from glass samples, g/m^2 ; C_i is the concentration of i element in solution, g/L ; f_i is the mass fraction of element i in the glass; SA/V is the ratio of glass surface area to the leachate volume, m^2/L .

The Raman spectra of the NBS5 glass powder were measured with a iHR550 device at room temperature. A 1200 line/mm grating was used, and a 532 nm semiconductor laser served as the excitation source. The Raman spectra were obtained in the standard range of $200\text{--}2000\text{ cm}^{-1}$ with a resolution of 1 cm^{-1} .

The Fourier-transform Infrared (FTIR) spectra of the borosilicate glasses were obtained using a PerkinElmer Spectrum II spectrophotometer in the attenuated total reflectance (ATR) mode. The wavenumber ranged from $450\text{ to }4000\text{ cm}^{-1}$ with a resolution of 2 cm^{-1} . The measured depth on the target surface by ATR-FTIR is less than $1.75\text{ }\mu\text{m}$ for a spectrum ranging larger than 1200 cm^{-1} [40].

The X-ray photoelectron spectroscopy (XPS) spectra of borosilicate glasses were acquired by the Thermofisher K-Alpha system. It uses a monochromatic and micro-focused Al $K\alpha$ beam with 1486.6 eV energy to excite the photoelectrons. The scanning depth of XPS spectra is 15 nm . A hemispherical analyzer collected the photoelectron spectra. The spectra range from $0\text{ to }1350\text{ eV}$, with a resolution of 0.5 eV (based on Ag $3d$ as a reference). The spectrum is typically acquired in $15\text{--}20\text{ min}$. For the full spectrum, a single sample is scanned once, while for the high-resolution spectrum, it is scanned five times.

The time-of-flight secondary ion mass spectrometry (ToF-SIMS, IONTOF GmbH) was employed to investigate the leaching profile of NBS5 glasses after leaching at a certain time. The samples were pumped overnight to ensure a 10^{-7} Pa vacuum level during the measurement. The Cs^+ ions, with an energy of 2 keV , were used for sputtering the surface. The sputtering area was set to $150 \times 150\text{ }\mu\text{m}^2$, and the analysis area was $40 \times 40\text{ }\mu\text{m}^2$. The analysis utilized a 25 keV monatomic Bi^+ beam ($\sim 1.0\text{ pA}$) operating at a 10 kHz frequency. Negative ions were selected for analysis, and a flood gun was used throughout to counteract any charging effect. Depths were gauged using a profilometer, correlating linearly with sputtering time [42].

3. Results

3.1. Acid environment leaching

3.1.1. Solution analysis

Fig. 1 illustrates the variations in normalized mass loss for silicon (Si), boron (B), and sodium (Na) in NBS5 glass, as a function of leaching time across different acidic environments. For Si, an increase in normalized mass loss is observed with rising pH levels. Despite this trend, the extent of Si release remains lower than that of elemental B and Na within the studied acidic range (pH 3–5). This pattern suggests a relatively minimal depolymerization of the Si network under these conditions. The analysis of elemental B reveals that its normalized mass loss is less pronounced when leached in a neutral pH environment (utilizing deionized water) than in acidic conditions. Notably, a significant escalation in normalized mass loss is observed under extremely acidic conditions (pH 3). Additionally, the leaching rate of elemental B at pH 4 exhibits a biphasic pattern: initially, the rate of change in normalized mass loss is substantially higher for leaching durations up to 8 h , following which the rate diminishes for prolonged leaching periods.

Similarly, the leaching dynamics of Na in acidic environments resemble those of B. The normalized mass loss of Na is markedly higher under severe acidic conditions (pH 3), with the rate of mass loss decreasing as the pH increases. Moreover, at pH 4, normalized mass loss of Na also demonstrates a variation in the rate of change over the leaching timeframe.

3.1.2. Microstructure evolution

The Attenuated Total Reflectance Fourier-transform Infrared (ATR-FTIR) spectra of NBS5 glass leached in deionized water (pH 6.5) are depicted in Fig. 2, which highlights several distinct regions that correspond to the various vibrational modes associated with different structures within the glass matrix. Each characteristic peak within the spectra provides insights into the complex bonding environment and the structural components that constitute the glass network. The peak at about 700 cm^{-1} is attributed to the tensile vibrational mode of the B-O-B [43,44], which is the basic building block of borosilicate glasses. The peak at about 780 cm^{-1} is indicative of the stretching vibrational mode of the oxy-silicon-oxygen unit (O-Si-O) [45]. For the representation of the Na element in FTIR spectra, it has been shown that the broad resonances in the range of $850\text{--}950\text{ cm}^{-1}$ are mainly contributed by the bonding of the Si network with Na^+ ions in the glass, particularly the resonance associated with the Si-O-2NBO (non-bridging oxygen) vibrational [46–48]. In addition, the peak at 945 cm^{-1} corresponds to

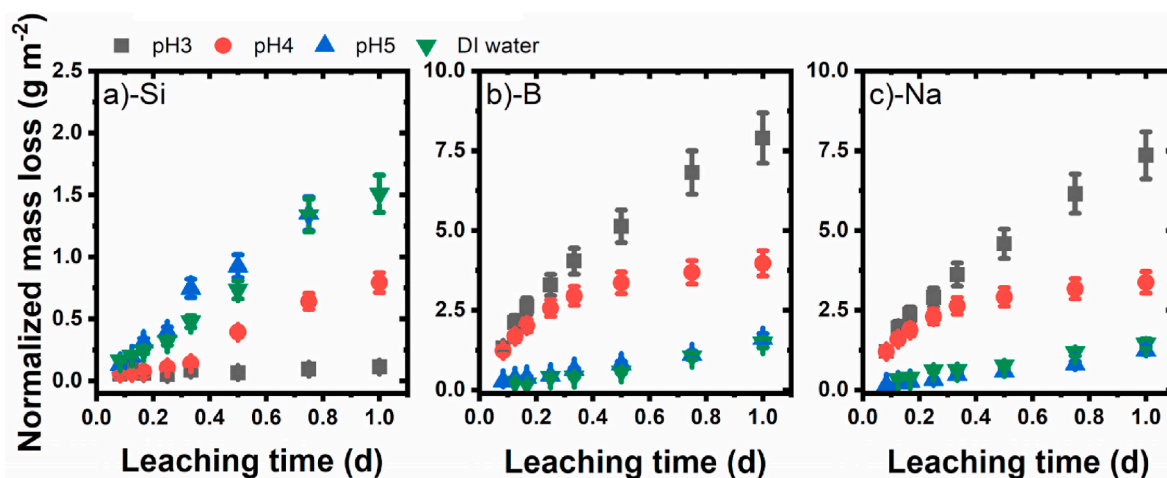


Fig. 1. Normalized elemental mass loss of NBS5 glass in different acidic environments. ((a), (b), and (c) represent the normalized mass loss of silicon (Si), boron (B), and sodium (Na), respectively.) Deionized water was used as a reference value with a pH of approximately 6.5. (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

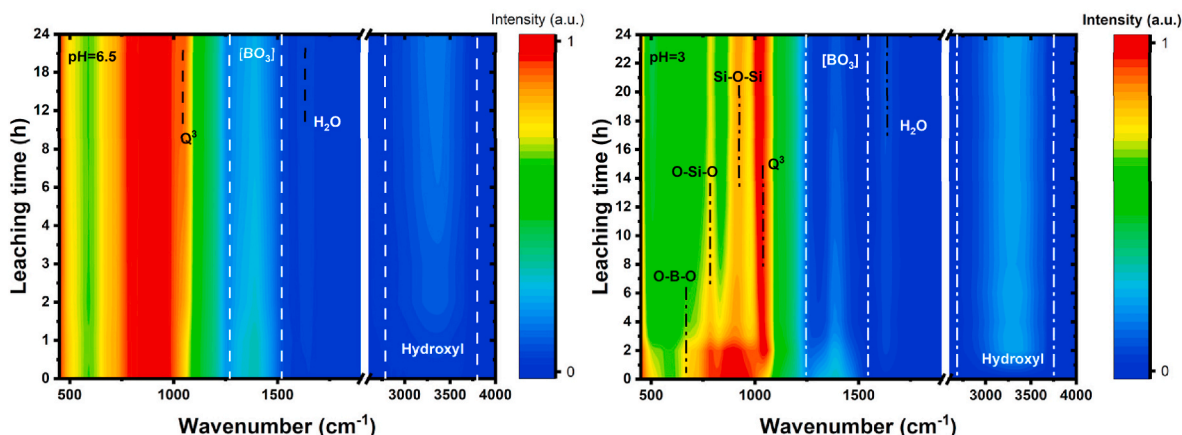


Fig. 2. The FTIR colormap of NBS5 glass after leaching in deionized water (pH 6.5, left) and pH 3 (right).

the stretching vibration of the borosilicate-oxygen bond (B-O-Si) [49], suggesting mixing between the borate and silicate networks. The peaks observed at 1050 cm^{-1} and 1150 cm^{-1} are related to the stretching vibrations of the silicon-oxy-silicon (Si-O-Si) bonds in the Q^3 and Q^4 structural units [49], which represent the degree of silicate network polymerization, with Q^4 indicating higher polymerization and networked structures. Finally, the stretching vibrations of the B-O bond in the $[BO_3]$ structural unit were observed in a wide range between 1250 and 1550 cm^{-1} .

After leaching, the microstructures of NBS5 glasses showed significant changes: including Si-related structural changes reflecting the independent appearance at 780 cm^{-1} , as well as more prominent fronts of the Q^3 structure. Furthermore, a significant decrease in the strength of the boron-related structural $[BO_3]$ unit located at 1250 – 1550 cm^{-1} [50]. For NBS5 glass samples after leaching, new structures appeared at 1633 cm^{-1} that should be associated with the H-O-H vibrations of molecular water [51,52], with which the FTIR absorption spectral shape typical of a large number of hydroxyl (O-H bending vibrations) interactions also appeared from 2600 to 3600 cm^{-1} [52].

Fig. 2 also showcases the FTIR colormap of NBS5 glass post-leaching in acidic conditions (pH 3), illustrating the pronounced evolution of the glass surface microstructure. The alteration is notably more severe in a strongly acidic environment (pH 3) than in neutral conditions (pH 6.5, as depicted in Fig. 2). Within the FTIR colormap analysis, significant modifications are observed in the glass network structures: the Q^3 structure, located at approximately 1050 cm^{-1} , and the Q^4 structure at 1150 cm^{-1} , undergo substantial changes, while the Si-related structure around 780 cm^{-1} appears entirely distinct.

Furthermore, the transformations in boron-related structures are markedly dramatic. The signal indicative of the $[BO_3]$ unit on the glass surface is almost entirely depleted through leaching, and the B-O-B structure, situated near 700 cm^{-1} , diminishes as leaching time increases. This is followed by alterations in the $[BO_4]$ structure, near 960 cm^{-1} , highlighting the extensive impact of acidic leaching on the glass's boron components.

The evolution of the surface microstructure of NBS5 samples after leaching in an acidic environment at pH 4 is shown in Supplementary Fig.A1. The overall evolution of the FTIR spectrum exhibits a reduction in intensity when compared to the infrared spectra of NBS5 samples subjected to leaching in a more severe acidic condition (pH 3). Nonetheless, the pattern of surface microstructural changes in the borosilicate glass at pH 4 mirrors those observed at pH 3, suggesting a consistency in the glass's response to acidic environments, albeit with a somewhat diminished effect at the slightly higher pH level. Moreover, the alterations in the NBS5 glass microstructure after leaching at pH 5 environment are similar to those observed in deionized water, albeit with significantly less intensity compared to the changes induced at pH 3 and

pH 4, which can be found in Supplementary Fig.A2. Notably, the evolution of the Q^3 structure, identifiable at 1050 cm^{-1} , is more distinctly pronounced on the sample surface. Moreover, while the evolution of boron-related structures persists, signals corresponding to the $[BO_3]$ unit are still detectable on the sample surface. This observation suggests that the leaching process has not completely penetrated the depths at which the FTIR $[BO_3]$ unit is located, indicating a shallower layer of alteration.

3.1.3. Leaching profile

Considering factors such as leaching depth and reaction rate, the sample tested for 6 h of leaching was chosen for atomic-level elemental analysis using XPS. A meticulous layer-by-layer sputtering approach facilitated the analysis of the sample's depth, focusing on a 500 nm range post-leaching. As shown in Fig. 3(a), the elemental analysis of NBS5 glass at 500 nm depth, with 100 nm step increments, after leaching at pH 3, reveals predominant signals for Si (Si 2p @ 103.44 eV) and O (O 1s @ 532.81 eV) in the surface layer. A weak Cl (Cl 2p @ 201.33 eV) signal detected in the topmost and bottom layers is hypothesized to be a residual trace from the leachate evaporation. Additionally, a notable C (C 1s @ 285.21 eV) signal, likely attributable to residual air contamination post-exposure, was observed. Crucially, 6 h of leaching process at pH 3, no Na or B signals were detected within the 500 nm surface range, suggesting the leaching layer had breached this depth. The XPS result of pH 4 is similar to that of pH 3 as shown in Supplementary Fig.A3.

Fig. 4 exhibits the B and Na distribution of NBS5 glass measured by ToF-SIMS after different leaching times at pH 3. The intensities of B and Na were normalized relative to the intensity of Si. Under acidic conditions, the B and Na interface depth increased significantly with leaching time, reaching approximately $1.4\text{ }\mu\text{m}$ after 6 h of leaching and about $3.6\text{ }\mu\text{m}$ after 24 h. This observation also confirms the XPS results that no signals of B and Na were detected within the 500 nm surface range.

3.2. Alkaline environment leaching

3.2.1. Solution analysis

The leaching dynamics of NBS5 glass under alkaline conditions were meticulously investigated to assess the stability of Si, B, and Na elements. Fig. 5 illustrates the normalized mass loss of these elements across a range of pH values (pH 9, 11, and 12), with deionized water serving as the baseline (pH 6.5). In stark contrast to neutral and mildly alkaline settings, a strongly alkaline environment markedly accelerates the leaching of Si, as depicted in Fig. 5-a. This pronounced leaching correlates with a substantial disruption of the Si network, particularly in environments with pH 11 and 12, significantly above neutral environments. Similar to Si, B and Na exhibit accelerated leaching rates in strongly alkaline environments. However, their leaching behaviors in

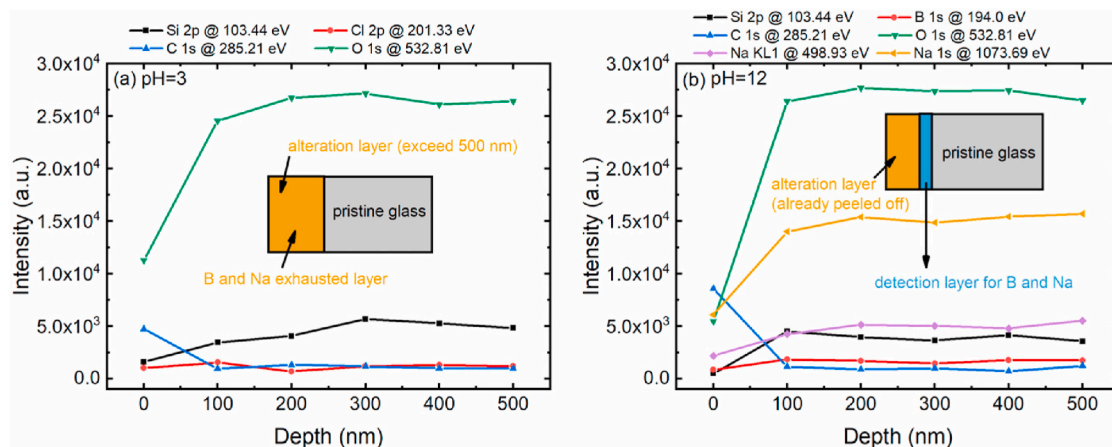


Fig. 3. XPS depth profile for NBS5 leaching at pH 3 (a) and pH 12 (b) with a leaching time of 6 h.

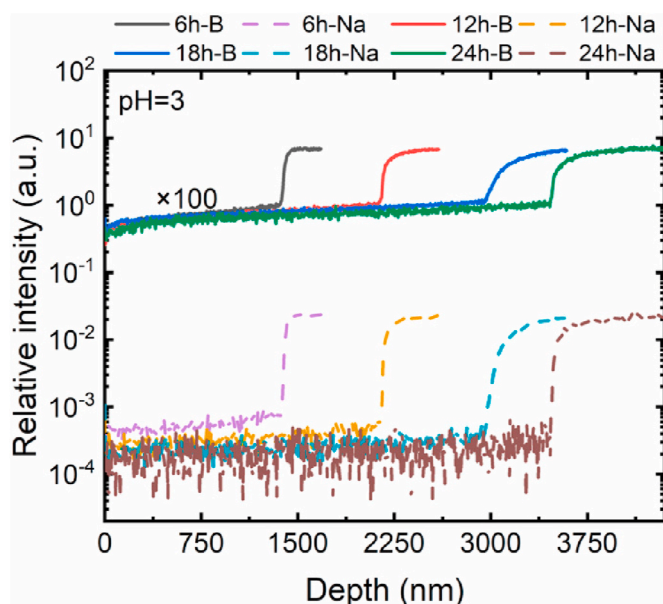


Fig. 4. Element distributions of B and Na over the depth for NBS5 glass after different leaching times at pH 3. (The intensities of B and Na were normalized relative to the intensity of Si).

mildly alkaline conditions (pH 9) closely align with those in neutral conditions, suggesting a lower degree of reactivity compared to more extreme pH levels.

3.2.2. Microstructure evolution

The ATR-FTIR spectral analysis provides insight into the microstructural evolution of borosilicate glass post-leaching. Fig. 6 shows the surface FTIR spectra of NBS5 samples after exposure to a pH 12 environment. The FTIR colormap indicates that the microstructural evolution remains subtle, underscoring the resilience of the glass network under highly alkaline conditions.

3.2.3. Leaching profile

Fig. 3(b) exhibits the details of the elemental composition of NBS5 glass after 6 h of leaching in pH 12 environment. The presence of Si and O signals dominates the surface layers, with C signals attributed to air exposure. Notably, the detection of B (B 1s @ 194.0 eV) and multiple Na (Na 2p @ 32.76 eV, Na KL1 @ 498.93 eV, and Na 1s @ 1073.69 eV) signals within the top 100 nm at pH 12 suggests a limited depth of

significant water reactivity, reinforcing the material's stability in alkaline conditions. The elemental analysis of NBS5 glass for 6 h after leaching in a pH 9 solution is shown in Supplementary Fig.A4. The overall trend is similar to leaching at pH 12, demonstrating the same leaching effects of an alkaline environment.

Fig. 7 shows the element distribution of NBS5 glass measured by ToF-SIMS after 18 h of leaching at pH 12. The intensities of other elements were normalized relative to the intensity of Si. It is noted that the interface thickness of B and Na remains nearly constant as the leaching time increases. The elemental distribution after 6 h and 12 h of leaching at pH 12 was displayed in Supplementary Fig.A5. The intensity of B and Na does not change with the depth increasing, only the interface depth of H slightly increases after 6 h of leaching. The elemental distribution after various times at pH 9 was depicted in Supplementary Fig.A6. The results indicate that in a weakly alkaline environment, the intensity of B does not exhibit a step change, suggesting that there is also no interface change. The statements mentioned above indicate that under an alkaline environment, B might not react with water directly, leading to a change in the B concentration in the glass network. Another possible description is that B and water would diffuse into each other in an alkaline environment; however, the layer where the interdiffusion occurred has already been detached into the leachate.

4. Discussion

4.1. Initial leaching rate

Table 2 presents the initial leaching rates (r_0) of key elements (Si, B, and Na) from borosilicate glass when subjected to leaching agents of various pH values. The initial leaching rate was derived from a linear regression analysis of the normalized mass loss of distinct elements with the leaching duration (within the first 24 h). Silicon, as the predominant constituent of borosilicate glass, demonstrates a noteworthy trend: its initial leaching rate increases with the pH, corresponding to a higher concentration of hydroxide (OH^-) ions. This observation underscores the limited alkali resistance of conventional borosilicate glass, highlighting its susceptibility to degradation in environments with elevated pH levels.

The interaction of borosilicate glass with water, particularly focusing on the elemental silicon and silicon-related structures, involves two key reaction mechanisms.

1. **Ion Exchange Reaction with Alkali Metals:** The presence of alkali metal ions (as network modifier, annotated as M^+) attached to the silicon-oxygen (Si-O) groups facilitates an ion exchange process with water, as represented by equation (2):

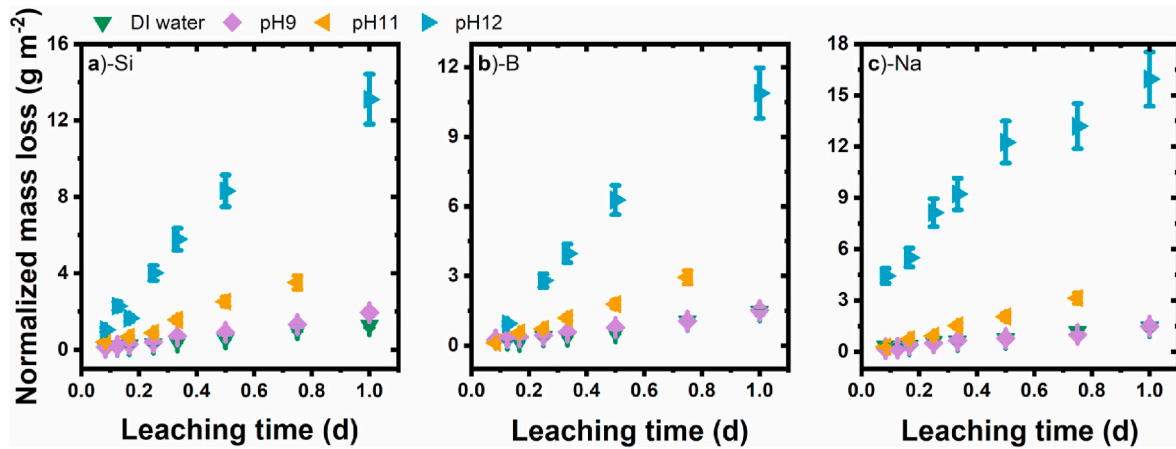


Fig. 5. Normalized elemental mass loss of NBS5 glass after leaching in different alkaline environments. ((a), (b), and (c) represent the normalized mass loss of silicon (Si), boron (B), and sodium (Na), respectively.) Deionized water was used as a reference value with a pH 6.5. (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

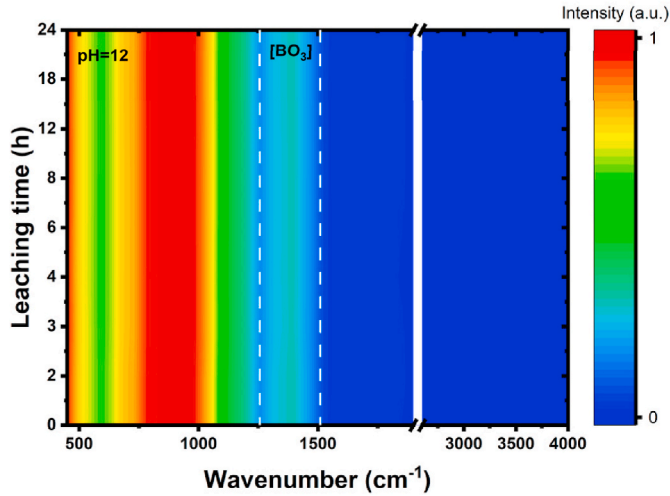
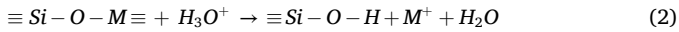
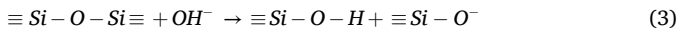


Fig. 6. Surface FTIR colormap of NBS5 glass after leaching in a pH 12 environment.



2. Silicon Network Hydrolysis: Silicon also reacts with hydroxyl (OH^-) ions present in the leach solution, a process that equation (3) describe:



The variation in the leaching rate of elemental Si in NBS5 glass under different pH conditions, as demonstrated in Fig. 8, provides critical insights into the impact of the leaching solution's pH on Si. Notably, the initial leaching rate of Si exhibits a significant decrease (approximately 9 times) in an acidic environment (pH 3) compared to leaching in neutral deionized water. This trend inversely correlates with pH levels; as the pH increases, the leaching rate of elemental Si becomes progressively more active, when pH 12, escalating to about 8 times the rate observed in neutral conditions. Such observations underscore the pronounced influence of OH^- ions from the leaching solution on silicon leaching behavior, whereas a higher concentration of H^+ ions appear to inhibit the leaching process of the silicon element.

Since the Si network will react with OH^- ions in the solvent, as the reaction process is shown in Equation (3), each Q^n group Si element to form $\equiv \text{Si} - \text{O} - \text{H}$ needs to consume a share of OH^- in the leaching agent,

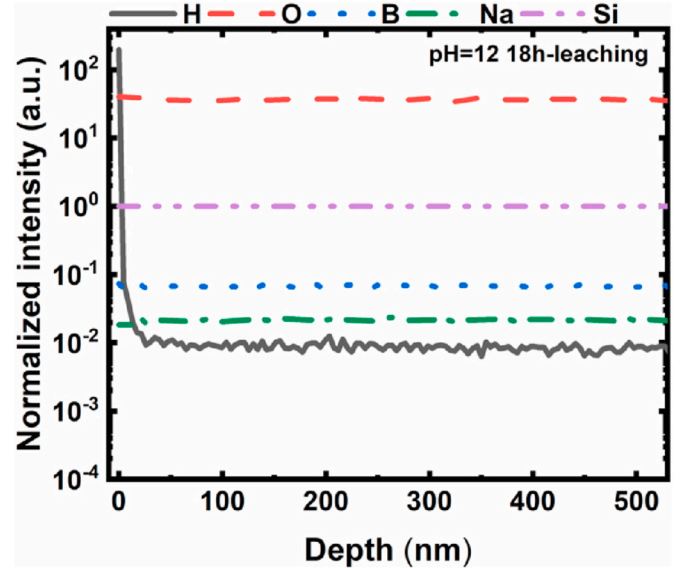


Fig. 7. Element distributions over the depth for NBS5 glass after 18 h of leaching at pH 12. (The intensities of other elements were normalized relative to the intensity of Si.)

Table 2

The initial leaching rate of main elements of NBS5 glass at various pH values.

pH value	$r_{\text{Si}} (\text{g} \cdot \text{m}^{-2} \cdot \text{d}^{-1})$	$r_{\text{B}} (\text{g} \cdot \text{m}^{-2} \cdot \text{d}^{-1})$	$r_{\text{Na}} (\text{g} \cdot \text{m}^{-2} \cdot \text{d}^{-1})$
3	0.2 ± 0.1	8.2 ± 0.8	7.4 ± 0.7
4	0.5 ± 0.1	3.5 ± 0.6	2.7 ± 0.5
5	1.7 ± 0.1	1.3 ± 0.1	1.1 ± 0.1
6.5	1.7 ± 0.1	1.3 ± 0.2	1.3 ± 0.1
9	1.8 ± 0.1	1.3 ± 0.1	1.5 ± 0.2
11	4.3 ± 0.5	4.1 ± 0.3	4.3 ± 0.3
12	12.0 ± 2.1	14.9 ± 0.9	14.0 ± 1.6

and the process will inevitably be due to increase in pH of leaching agent. Assuming the vast majority of the newly generated $\equiv \text{Si} - \text{O} - \text{H}$ goes into solution, then the amount of $\equiv \text{Si} - \text{O} - \text{H}$ would increase after 2 h of leaching at different initial pH values.

Gin et al. [19] examined the leaching behavior of CJ1 glass (ternary borosilicate sodium) at pH 9 and 90°C. They found that the initial leaching rate of silicon was $11.6 \pm 3.5 \text{ g}/(\text{m}^2 \cdot \text{d})$, which is about 7–8

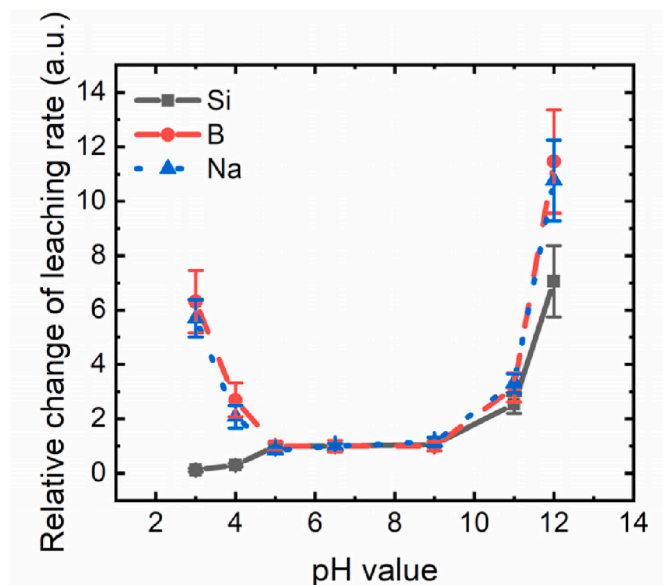


Fig. 8. The relative change of initial leaching rate of elements with the pH value of leachate. (All initial leaching rates have been normalized based on the corresponding values under neutral conditions (pH 6.5)).

times higher than the results reported in this study. This discrepancy is attributed to the fact that they prepared the pH 9 at 90°C, whereas we prepared the pH at room temperature. When the leaching solution is heated up to 90°C, the pH value subsequently decreases. Furthermore, Gin et al. [19] conducted their leaching study using powder, resulting in an equivalent specific surface area that was three times greater than that used in this study. This increase in specific surface area significantly accelerates the leaching kinetics of the glass.

After leaching, the initial leaching rate of elemental B from NBS5 glass, as inferred from the mass loss data, exhibits a U-shaped curve, as illustrated in Fig. 8. Specifically, this U-shaped trend indicates that the migration rate of elemental B into the leaching solvent intensifies under conditions of strong acidity and alkalinity. Conversely, in environments that are weakly acidic, weakly alkaline, or neutral, the reaction rate stabilizes, demonstrating a more consistent leaching behavior.

Notably, the leaching rate of elemental B is observed to be faster in a strongly alkaline solution (pH 12) than in a strongly acidic solution (pH 3). This observation suggests that the depolymerization of the borosilicate glass network is more pronounced in the presence of OH⁻ ions-rich solvents. This phenomenon underscores the significant impact of the chemical nature of the leaching environment on the structural integrity and chemical stability of borosilicate glass, highlighting the critical role of pH in influencing the leaching dynamics of glass constituents.

Similarly, the initial leaching rate of Na is influenced by the concentration of H⁺ and OH⁻ ions, with reduced rates observed at pH 5 and pH 9 compared to deionized water. This discrepancy may stem from variations in standard concentration solutions across different batches. Despite expectations of higher Na ion leaching rates in acidic environments, ICP-OES testing reveals a more than 10 times increase in initial Na ion leaching at pH 12 compared to deionized water, suggesting Na loss accompanies the disintegration of the borosilicate network.

4.2. Leachate analysis

The pH measurement of NBS5 leaching solutions post-reaction reveals distinct behaviors across different initial pH values, as depicted in Fig.A.7. In strongly acidic conditions (pH 3), the solution's pH exhibits minimal change, consistently remaining at pH 3 throughout the leaching process. Conversely, strong alkaline solutions (pH 12) demonstrate a slight decrease in pH to approximately 11.8 after 24 h, suggesting the

consumption of OH⁻ ions during the glass-water interaction. Weakly acidic solutions exhibit a gradual increase in pH, whereas, at pH 9, a significant reduction in OH⁻ ion concentration occurs, resulting in a pH decrease to approximately 6.5, indicating a substantial alteration in solution chemistry.

Table 3 outlines the pH value of the leachate 2 h post-reaction for different initial pH values, and includes the deduced concentrations of H⁺ and OH⁻ ions of each initial pH value. The findings indicate a minimal release of elemental Si within the initial 2 h of reaction with water in acidic conditions. This minimal activity can be attributed to the lower probability of OH⁻ ions reacting with the Si network structure, as dictated by Equation (3), owing to the reduced concentration of OH⁻ ions in such environments.

At the start of the leaching experiment at pH 9, the detection of 0.14 mg/L of elemental Si in the leach solution after 2 h reaction period signifies the precipitation of approximately 5.04 µg Si triggered by the water-glass reaction. According to Equation (3), the precipitation of each portion of elemental Si necessitates the binding of an equivalent portion of OH⁻ ions, therefore affecting the pH value of the leaching solution. Theoretically, this reaction would adjust the leach solution's pH to approximately 8.7, assuming only the silicon-binding OH⁻ ions contribute to pH changes. Contrary to theoretical expectations, the pH of the leach solution was observed to decrease to about 6.5 after the 2-h reaction, as depicted in Supplementary Fig.A.7. This discrepancy suggests a more extensive reaction according to Equation (3), leading to the formation of ≡ Si-O-H structures. A significant portion of these reaction products remains at the glass-water interface, contributing to the development of a gel layer that impedes further leaching.

The XPS and ToF-SIMS findings (as described in Figs. 3 and 7) suggest that the water-resistant layer formed after the leaching process is thinner, not exceeding 100 nm. To better verify the formation of the ≡ Si-O-H structure during the water-glass interaction in alkaline leachate, the experimental NBS5 powder was leached at pH 9 and the 2 h-leaching solution was characterized using Raman spectroscopy, and the results are shown in Supplementary Fig.A8. It can be found that the Raman spectra of the leached powder samples show an obvious sharp signal at about 560 cm⁻¹ and about 1550 cm⁻¹, which directly corresponds to the signal structure of the original silicic acid (H₂SiO₃) [53,54]. Meanwhile, in the Raman spectrum, the Si-OH-like structure and Qⁿ structure associated with Na-Si and B elements still exist, which indicates that the main body of the leached powder sample is still an amorphous glassy structure. Another obvious change in the Raman spectra of the leached NBS5 powder sample is in the region of 1300–1700 cm⁻¹, and the Raman spectra of the silicic acid in the corresponding region have two obvious signals located at 1421 cm⁻¹ and 1549 cm⁻¹, and the ratio of the relative intensities of these two peaks is consistent with the ratio of the signal intensity of the peaks in the powder sample of NSB5, which also indicates the existence of Si-OH structure.

4.3. Microstructural changes

Analyzing the FTIR spectra for microstructural changes, the spectra

Table 3

Concentration of elemental Si in the solution after leaching and concentration of H⁺ and OH⁻ ions in the initial leachate.

pH value	NI _{Si-2h} (mg·L ⁻¹)	Number (Si)	pH value 2 h leaching	Initial (H ⁺)/ions	Initial (OH ⁻)/ions
3	0.05	3.74×10^{16}	2.96	2.17×10^{19}	2.17×10^{11}
4	0.06	4.48×10^{16}	4.13	2.17×10^{18}	2.17×10^{12}
5	0.13	9.72×10^{16}	5.81	2.17×10^{17}	2.17×10^{13}
6.5	0.16	1.19×10^{17}	–	6.86×10^{15}	6.86×10^{14}
9	0.14	1.05×10^{17}	6.43	2.17×10^{13}	2.17×10^{17}
11	0.41	3.06×10^{17}	–	2.17×10^{11}	2.17×10^{19}
12	1.05	7.85×10^{17}	11.96	2.17×10^{10}	2.17×10^{20}

were normalized using specific troughs at approximately 580 cm^{-1} as reference points for structural signal absence. Supplementary Fig.A9 illustrates the differential FTIR spectra of NBS5 glass post-leaching in a pH 3 environment, where the baseline is the original sample spectrum and each leached sample's spectrum is subtracted from the baseline. Post-leaching, an increase in signal intensity is observed for Si-related Q^3 and Q^4 structures, alongside water-related structures at 1633 cm^{-1} . However, a notable increase in signal intensity for boron-related structure ($[BO_4]$ near 700 cm^{-1} , at 960 cm^{-1} , and the $[BO_3]$ structural unit at $1250\text{--}1550\text{ cm}^{-1}$) is evident, highlighting accelerated microstructural evolution in acidic conditions compared to neutral environments. This is further supported by a reduction in signal intensity at approximately 500 cm^{-1} , attributed to Si-O-Si and O-Si-O bending vibrations within the Q^4 unit and B-O-B bonds [55], underscoring significant alterations in the glass matrix.

After analyzing the time-dependent difference-by-difference spectrum of borosilicate glass after-leaching in pH 3 environment, attention shifts to examining the leaching dynamics of elemental B. This examination begins with an in-depth analysis of the $[BO_3]$ region through peak fitting techniques to observe the changes in structural intensity over time, as depicted in Fig. 9. Notably, the intensity of the $[BO_3]$ unit on the surface of the sample exhibits a rapid decrease in an acidic setting, eventually stabilizing at a saturation point. This saturation is attributed primarily to the depletion of $[BO_3]$ units within the water corrosion layer [40], demonstrating a significant alteration in the chemical composition of the glass surface under acidic environments.

In neutral and acidic environments, the $[BO_3]$ unit depletion rate in NBS5 glasses exhibits a direct correlation with the square root of time, indicative of the solid-liquid diffusion of $[BO_3]$ unit during the glass-water reaction. This rate flattens as pH increases, suggesting that a higher concentration of OH^- ions moderately inhibits the solid-liquid diffusion of $[BO_3]$ unit.

In the context of acidic leaching of NBS5 glass, the leaching dynamics are characterized by a swift rate, leading to the surface $[BO_3]$ structure achieving equilibrium at pH 3 about 4 h after leaching (as evidenced by the data presented in Supplementary Fig.A9). This equilibrium suggests that the $[BO_3]$ structure within the leached layer is nearly fully dissolved into the solution within this timeframe. Consequently, an analysis of the $[BO_3]$ leaching behavior in acidic environments necessitates focusing on the data from the initial 4 h of leaching to accurately model the relative rate of change.

Fig. 9 illustrates the relationship between molecular water content and the square root of leaching time across various pH conditions. Under

alkaline conditions, the intensity of molecular water is minimal and remains constant over leaching time. In weakly acidic and neutral environments, a linear correlation exists between molecular water content and the square root of leaching time. Under strongly acidic conditions (pH 3 and 4), a robust linear relationship persists, with the intensity of molecular water exhibiting a significant increase. This suggests that leaching behavior under acidic and neutral conditions is primarily governed by interdiffusion [36], with the reaction rate positively correlated to the pH of solutions.

The appearance of molecular water-related structures on the glass surface post-leaching serves as direct evidence of solid-liquid interdiffusion, with a nearly perfect linear relationship observed between water-related structure changes and $[BO_3]$ unit depletion (as shown in Supplementary Fig.A10), excluding leaching at pH 3 and pH 4. This relationship underscores the synchronized process of $[BO_3]$ unit leaching and water entry into the glass surface layer, facilitated by the microporous structure left behind by departing $[BO_3]$ units. The analysis reveals a distinct pattern in the interaction between water and the borosilicate glass surface, particularly regarding the $[BO_3]$ structural units. Apart from the outlier case at pH 3 and pH 4, a remarkably linear correlation exists between the leaching of $[BO_3]$ unit and the incorporation of water into the glass surface layer. This correlation suggests that the processes of $[BO_3]$ unit leaching and water penetration are synchronized, occurring concurrently during the leaching reaction. The mechanism underlying this synchronicity involves solid-liquid diffusion, where the $[BO_3]$ units are leached from the glass surface, forming numerous micropores. These micropores facilitate the penetration of molecular water, allowing it to comprehensively infiltrate and saturate the glass surface's water-etched layer. This phenomenon highlights the dynamic interplay between the structural changes in the glass due to leaching and the consequential modification of its surface porosity and hydration state.

The comparison of the linear relationship between the $[BO_3]$ unit and water (signified by the 1633 cm^{-1}) on the surface of the sample

Table 4

Linear fitting parameters of the $[BO_3]$ relative change and area of water peaks on the glass surface.

pH value	Slope	Intercept	R^2
3	6.66 ± 0.12	0.60 ± 0.06	0.999
4	4.71 ± 1.20	1.25 ± 0.83	0.877
Others	5.19 ± 0.29	0.03 ± 0.01	0.982

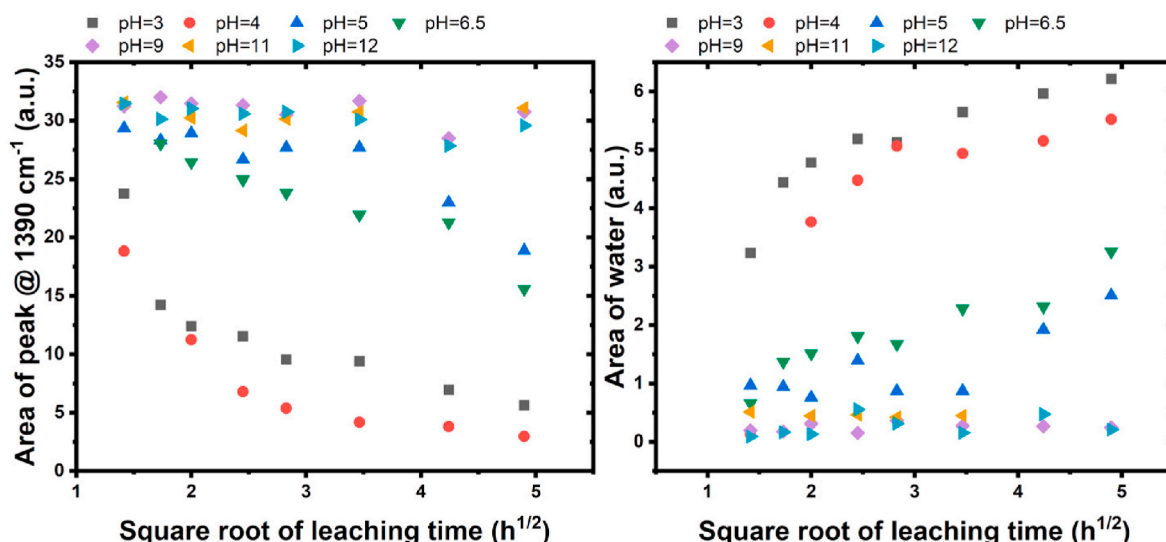


Fig. 9. $[BO_3]$ structure units (left) and molecular water (right) deconvolution fitting in NBS5 varying with the square root of leaching time.

post-leaching under pH 3 and pH 4 conditions versus other scenarios indicates higher slopes (as presented in Table 4). This difference underscores a robust correlation between the water-related structures on the sample surface and the $[\text{BO}_3]$ units, indicating a constant mechanism of interaction at different pH values, with the exception of a more noticeable reaction progress at pH 3 and pH 4. The observed discrepancy at pH 3 and 4, characterized by distinct intercepts (as presented in Table 4), may be attributed to the specific ionic interactions prevalent in strongly acidic environments. It's hypothesized that the presence of Na^+ ions, which serve as charge-compensating entities around the $[\text{BO}_4]$ unit, undergo extensive ion exchange processes with H^+ ions in the solution. This exchange likely facilitates the transformation of a significant fraction of $[\text{BO}_4]$ units into $[\text{BO}_3]$ units as an initial reaction step. This phenomenon is proved by Aréna et al. [56] using electron energy loss (EEL). Such a transformation would not only alter the structural composition of the glass surface but also impact the leaching dynamics and the subsequent formation of water-related structures. This complex series of reactions and structural transformations is illustrated in Supplementary Fig.A11 leaching behavior at pH 3 and 4. Gin et al. [19] pointed out that the fraction of $[\text{BO}_4]$ in CJ1 glass (ternary borosilicate glass) is 66 %, and the composition of NBS5 glass is similar to CJ1 glass, a higher content of $[\text{BO}_4]$ would exacerbate the transformation.

4.4. The leaching behavior of surface elements

Fig. 10 presents XPS spectra comparing the surface elements of NBS5 glass after 6 h of leaching across different pH environments. Normalization was performed against the C 1s peaks to account for similar carbon surface contamination across samples. The findings indicate comparable Si signal intensities and Si/O signal intensity ratios in both acidic and weakly alkaline environments, suggesting uniformity in surface Si/O fractions. However, in alkaline conditions, a significant decrease in the Si/O intensity ratio is observed, aligning with the introduction of new structures and a substantial drop in surface Si elemental intensity, especially at pH 12. This corresponds with increased Si element mass loss in the solvent post-leaching, indicating enhanced reactivity and structural modification in strongly alkaline environments.

It can be found that for both acidic and weakly alkaline environments, the Si signal intensity on the surface of the samples is around the same magnitude (e.g., Fig. 10-a), and the Si/O signal intensity ratio is similar in acidic environments (e.g., Fig. 10-b). This indicates the similarity of their surface Si/O fractions. However, Si/O intensity is drastically decreased for the alkaline environment compared to the acidic environment, probably due to the emergence of new structures. It can also be noticed that the Si elemental intensity on the surface of the samples is very low after leaching at pH 12, which also corresponds to a

significant increase in the mass loss of Si elements measured in the solvent after leaching.

4.5. The interface depth of boron in different pH conditions

Table 5 presents the interface depth of B, as measured by ToF-SIMS, after leaching under different pH conditions. The ToF-SIMS results after various leaching times at pH 6.5 are presented in Supplementary Fig. A12. The interface depth of B at pH 3 is approximately 10 times greater than at pH 6.5, indicating that B reacts more strongly under acidic conditions. This is consistent with the observed change in the initial boron leaching rate (r_0) under the pH 3 and pH 6.5 conditions (as presented in Table 2). The initial leaching rate of B at pH 3 is approximately 6 times that at pH 6.5. Gin et al. [57] reported that the thickness of the altered layer in glass could be determined by dividing the normalized mass loss of B into the glass density. The B interface depths determined by ICP-OES at various pH conditions are shown in Table 5. Fig. 11 showcases the discrepancy in the B interface depth between the ICP-OES and ToF-SIMS. In the neutral condition (pH 6.5), the B interface depth calculated by ICP-OES is deeper than that of ToF-SIMS. However, in a strong acidic level (pH 3), the B interface depths measured by the two methods are approximately identical within the error range. This phenomenon indicated that the main leaching mechanism between B and the leaching solution in pH 3 might be interdiffusion. In the neutral condition, the ICP-OES B interface depth is deeper than that of ToF-SIMS, the reason might be that the leaching mechanism is controlled by interdiffusion and network hydrolysis. The ToF-SIMS B interface depth is only the depth for interdiffusion, when the measurement has proceeded, a certain thickness of the altered layer has been peeled into the leachate. However, this process would not affect the B interface depth calculated by ICP-OES, which leads to the ToF-SIMS boron interface depth lower than that of ICP-OES. The discrepancy

Table 5

The boron interface depth determined by ToF-SIMS and ICP-OES after various leaching times at different pH conditions.

pH value	Interface depth of boron (nm)							
	6 h		12 h		18 h		24 h	
	ToF	ICP	ToF	ICP	ToF	ICP	ToF	ICP
3	1401	1352	2192	2103	3123	2795	3567	3239
DI water (~6.5)	159	238	228	352	322	608	397	684
12	43 ^a	1150	44 ^a	2574	43 ^a	–	43 ^a	4461

^a This signal represents the corresponding depth of the 10 times acquisition signal time. Data points within this range might be unreliable due to surface effects.

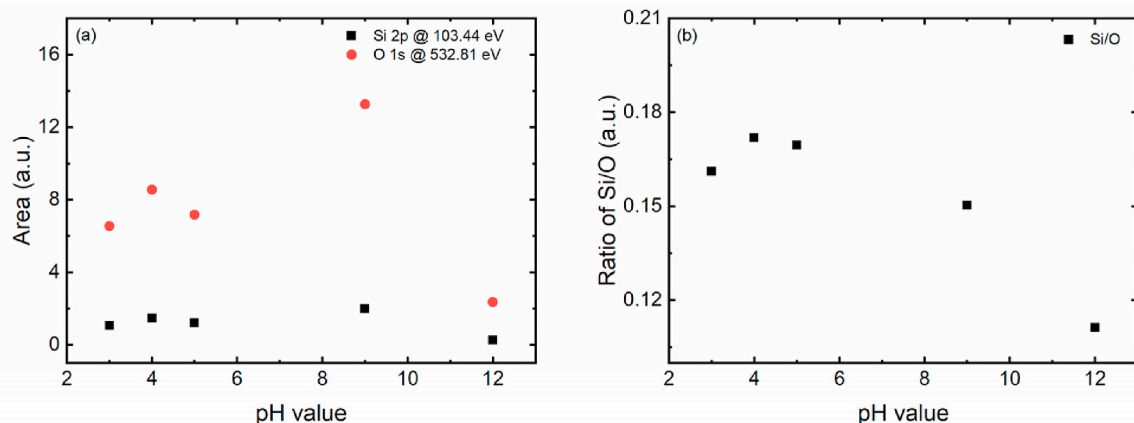


Fig. 10. a) Area of Si 2p and O 1s signal of XPS spectrum of NBS5 glass surface after 6 h leaching at different pH leachate; b) the ratio of Si/O signal area.

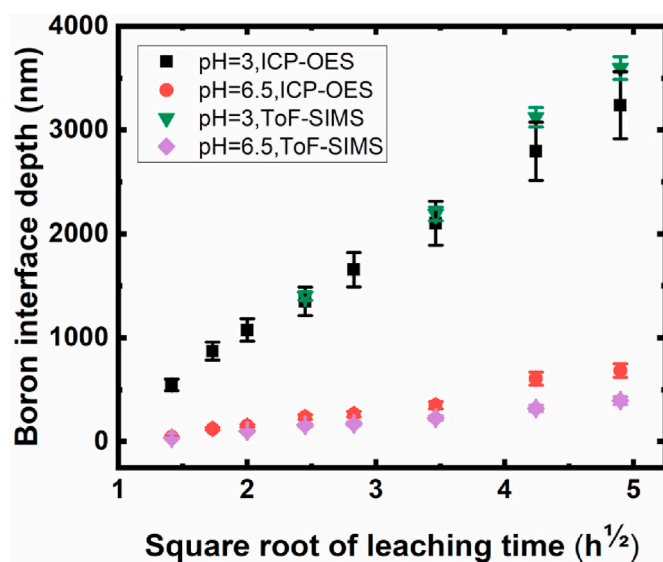


Fig. 11. Boron interface depth determined by ICP-OES and ToF-SIMS over square root of leaching time.

between these two methods might be the thickness of the altered layer peeled into the leachate.

Although the observation in ToF-SIMS did not reveal the B interface depth at pH 12, the initial leaching rate of B, Na, and Si at pH 12 increased approximately 10 times compared to pH 6.5 (as presented in Table 2). This implies that the glass network was destroyed and elemental B, Na, and Si were released into the leachate. However, FTIR and XPS results indicate that the microstructure and elemental distribution of the leached glass at pH 12 are similar to that of the pristine glass. The phenomenon mentioned above implies that the corrosion of NBS5 glass in alkaline conditions might be caused by peeling layer by layer. It is noted that the B interface depth at pH 12 calculated by ICP-OES (as depicted in Table 5) is even deeper than that in a strong acidic condition (pH 3). However, these depths do not mean the actual thickness of the altered layer under alkaline conditions (pH 12).

Based on the above results, the leaching behavior of the NBS5 glass in various pH conditions could be summarized. The main reaction mechanism under acidic conditions is interdiffusion, which includes solid-liquid interdiffusion between B and water, and ion exchange between Na^+ and $\text{H}^+/\text{H}_3\text{O}^+$ which processes mainly occur in B and Na elements. In the neutral condition (deionized water), the controlled mechanism between glass and water is interdiffusion and network hydrolysis. While under alkaline conditions, the main reaction-controlling mechanism becomes the dissolution of the glass network structures, leading to the glass peeled layer by layer and the main glass elements (B, Na, and Si) dissolving into the leachate [58]. A. H. Mir also observed a similar phenomenon [59].

5. Conclusion

In this study, analytically pure HCl and electronic grade KOH were meticulously employed to prepare leaching solutions across a spectrum of pH levels (pH 3–12), the leaching experiments were conducted with the MCC-1 static leaching method at 90°C, aiming to elucidate the leaching behaviors of Si, B, and Na within ternary sodium borosilicate glasses. After the leaching experiments, ICP-OES measured the Si, B, and Na concentrations in the leachate. FTIR analyzed the evolution of the microstructures of pristine and post-leaching samples. The leaching profiles were obtained using the XPS and ToF-SIMS. These results revealed different Si, B, and Na fates in the leaching process.

Silicon, the primary network-forming element in borosilicate glasses, exhibits differential leaching resistance across various compositions,

however, it maintains consistent overall leaching kinetics. Notably, the Si network structure demonstrates robust acid resistance, preserving its integrity across acidic, neutral, and mildly alkaline environments (pH 9). Under these conditions, the initial leaching rate of Si remains consistently low. However, in strongly alkaline conditions (pH 11 and pH 12), the leaching rate of Si markedly increases, significantly surpassing that in deionized water. This phenomenon leads to the disintegration of the borosilicate glass network by the elevated concentration of OH^- ions, thus heightening the risk of radionuclide leakage from high-level radioactive waste.

The leaching dynamics of elemental B are predominantly governed by solid-liquid diffusion except for leaching in an alkaline solution. It is observed that the initial leaching rate of B follows a U-shaped curve, indicating accelerated disintegration of the B network structure in both strongly acidic and alkaline environments. The leaching mechanisms for elemental B diverge significantly under environments rich in H^+ versus OH^- ions. In acidic conditions, the prevalent $\text{H}^+/\text{H}_3\text{O}^+$ ions might facilitate ion exchange with the network-modifying alkali metal element Na, further leading to a transformation of $[\text{BO}_4]$ units into $[\text{BO}_3]$ units. This transformation promotes solid-liquid interdiffusion of $[\text{BO}_3]$, culminating in the substantial release of B into the solution. Conversely, in alkaline environments, the depolymerization of the Si network results in rapid microstructural degradation and an increased leaching rate of elemental B.

The leaching behavior of Na mirrors that of B, heavily influenced by ion exchange dynamics within the solution. The acidic environment significantly boosts the leaching behavior of Na, with its leaching behavior in alkaline conditions attributed to the comprehensive disintegration of the glass network structures.

This investigation provides a comprehensive analysis of the leaching behaviors of key elements in borosilicate glass, shedding light on the intricate interplay between glass composition, environmental pH, and the resultant leaching dynamics. However, the composition of nuclear waste glass is intricate, and its leaching behavior is consequently rendered even more intricate. The doping of various elements and the incorporation of radioactive waste may lead to remarkable modifications in nuclear waste glass the leaching dynamics.

CRedit authorship contribution statement

Kai Bai: Writing – review & editing, Writing – original draft, Conceptualization. **Kemian Qin:** Writing – review & editing, Writing – original draft, Methodology. **Zhaoxuan Jin:** Writing – review & editing. **Buyun Zhang:** Writing – review & editing. **Yuchuan Wang:** Data curation. **Huashuai Hu:** Writing – review & editing. **Ahsan Ejaz:** Writing – review & editing. **Peng Lv:** Funding acquisition. **Tieshan Wang:** Funding acquisition. **Haibo Peng:** Writing – review & editing, Supervision, Funding acquisition.

Data availability

The data used in this study is available from the corresponding author upon reasonable request.

Declaration of interests

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

We express our sincere gratitude to the reviewers for their constructive comments. This work was supported by the National Natural Science Foundation of China (Grant No. 12175092), the Special Funds for Basic Scientific Research Operation Expenses of Central

Universities, Lanzhou University (Grant No. LZUJBKY-2023-STLT01), and the Natural Science Foundation of Gansu province (Grant No. 23JRRA1042, Grant No. 23JRRA1145). Haibo Peng was supported by INWARD coordinated research project [Ion Beam Irradiation for High-Level Nuclear Waste Form Development, F11022] from International Atomic Energy Agency (IAEA).

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.ceramint.2025.01.352>.

References

- [1] R.C. Ewing, W.J. Weber, F.W.C. JR, Radiation effects in nuclear waste forms for high-level radioactive waste, *Prog. Nucl. Energy* 29 (2) (1995) 63–127.
- [2] W.J. Weber, Radiation and thermal ageing of nuclear waste glass, *Procedia Materials Science* 7 (2014) 237–246.
- [3] S. Peugeot, P.Y. Noël, J.L. Loubet, et al., Effects of deposited nuclear and electronic energy on the hardness of R7T7-type containment glass, *Nucl. Instrum. Methods Phys. Res.* 246 (2) (2006) 379–386.
- [4] S.V. Raman, Microstructures and leach rates of glass ceramic nuclear waste forms developed by partial vitrification in a hot isostatic press, *JMatS* 33 (1998) 9.
- [5] V.I. Malkovsky, S.V. Yudin, M.I. Ojovan, et al., The influence of radiation on confinement properties of nuclear waste glasses, *Science and Technology of Nuclear Installations* (2020) 2020.
- [6] W.G. Burns, A.E.H. J.A.C. Marples, R.S. Nelson, A.M. Stoneham, Radiation effect and leaching rate of vitrified radioactive waste, *Natur* 295 (1982) 3.
- [7] S. Gin, A. Abdelouas, L.J. Criscenti, et al., An international initiative on long-term behavior of high-level nuclear waste glass, *Mater. Today* 16 (6) (2013) 243–248.
- [8] S. Gin, P. Jollivet, M. Tribet, et al., Radionuclides containment in nuclear glasses: an overview, *Radiochim. Acta* 105 (11) (2017) 927–959.
- [9] B.Y. Zubekhina, B.E. Burakov, M.I. Ojovan, Surface alteration of borosilicate and phosphate nuclear waste glasses by hydration and irradiation, *Challenges* 11 (2) (2020) 14.
- [10] A.H. Mir, A. Jan, J.-M. Delaye, et al., Effect of decades of corrosion on the microstructure of altered glasses and their radiation stability, *npj Mater. Degrad.* 4 (1) (2020) 1–9.
- [11] D.M. Strachan, R.P. Turcotte, B.O. Barnes, MCC-1: a standard leach test for nuclear waste forms, *Nucl. Technol.* 56 (2) (2017) 306–312.
- [12] K. Bai, X. Chen, J. Mao, et al., Coupling effects in borosilicate glass leaching: a study on La/V doping, *J. Nucl. Mater.* (2024) 593.
- [13] C.-W. Kim, J.-K. Park, T.-W. Hwang, Analysis of leaching behavior of simulated LILW glasses by using the MCC-1 test method, *J. Nucl. Sci. Technol.* 48 (7) (2011) 1108–1114.
- [14] O.M. Farid, M.I. Ojovan, A. Massoud, et al., An assessment of initial leaching characteristics of alkali-borosilicate glasses for nuclear waste immobilization, *Materials* 12 (9) (2019).
- [15] T.L. GötT, M.T. Harrison, I. Farnan, Evaluating the temperature dependence of Magnox waste glass dissolution, *J. Non-Cryst. Solids* 518 (2019) 75–84.
- [16] M. Fournier, A. Ull, E. Nicoleau, et al., Glass dissolution rate measurement and calculation revisited, *J. Nucl. Mater.* 476 (2016) 140–154.
- [17] J. Neeway, A. Abdelouas, B. Grambow, et al., Vapor hydration of SON68 glass from 90°C to 200°C: a kinetic study and corrosion products investigation, *J. Non-Cryst. Solids* 358 (21) (2012) 2894–2905.
- [18] C. Jegou, S. Narayanasamy, F. Angeli, Short communication on the Influence of the temperature between 30 and 70°C on the hydration of SON68 nuclear waste glass in a vapour phase, *J. Nucl. Mater.* (2021) 545.
- [19] S. Gin, A.H. Mir, A. Jan, et al., A general mechanism for gel layer formation on borosilicate glass under aqueous corrosion, *J. Phys. Chem. C* 124 (9) (2020) 5132–5144.
- [20] R. Hellmann, S. Cotte, E. Cadel, et al., Nanometre-scale evidence for interfacial dissolution-reprecipitation control of silicate glass corrosion, *Nat. Mater.* 14 (3) (2015) 307–311.
- [21] M. Tribet, A.H. Mir, C. Gillet, et al., New insights about the importance of the alteration layer/glass interface, *J. Phys. Chem. C* 124 (18) (2020) 10032–10044.
- [22] S. Gin, C. Guittonneau, N. Godon, et al., Nuclear glass durability: new insight into alteration layer properties, *J. Phys. Chem. C* 115 (38) (2011) 18696–18706.
- [23] P. Frugier, S. Gin, Y. Minet, et al., SON68 nuclear glass dissolution kinetics: current state of knowledge and basis of the new GRAAL model, *J. Nucl. Mater.* 380 (1–3) (2008) 8–21.
- [24] S. Gin, P. Jollivet, M. Fournier, et al., Origin and consequences of silicate glass passivation by surface layers, *Nat. Commun.* 6 (2015) 6360.
- [25] S. Gin, X. Beaudoux, F. Angeli, et al., Effect of composition on the short-term and long-term dissolution rates of ten borosilicate glasses of increasing complexity from 3 to 30 oxides, *J. Non-Cryst. Solids* 358 (18–19) (2012) 2559–2570.
- [26] K. Qin, X. Zhang, K. Bai, et al., Composition-dependent leaching behavior in ion-irradiated borosilicate glasses: a corrosion mechanism study, *J. Am. Ceram. Soc.* 107 (2) (2023) 859–870.
- [27] R. Bouakkaz, A. Abdelouas, B. Grambow, Kinetic study and structural evolution of SON68 nuclear waste glass altered from 35 to 125 °C under unsaturated H₂O and D₂O vapour conditions, *Corrosion Sci.* 134 (2018) 1–16.
- [28] E.M. Pierce, E.A. Rodriguez, L.J. Calligan, et al., An experimental study of the dissolution rates of simulated aluminoborosilicate waste glasses as a function of pH and temperature under dilute conditions, *Appl. Geochem.* 23 (9) (2008) 2559–2573.
- [29] T. De Echave, M. Tribet, P. Jollivet, et al., Effect of clayey groundwater on the dissolution rate of SON68 simulated nuclear waste glass at 70 °C, *J. Nucl. Mater.* 503 (2018) 279–289.
- [30] P. Jollivet, S. Gin, S. Schumacher, Forward dissolution rate of silicate glasses of nuclear interest in clay-equilibrated groundwater, *Chem. Geol.* 330–331 (2012) 207–217.
- [31] S. Gin, J.P. Mestre, SON 68 nuclear glass alteration kinetics between pH 7 and pH 11.5, *J. Nucl. Mater.* 295 (2001) 83–96.
- [32] M. Fournier, T. Ducasse, A. Pérez, et al., Effect of pH on the stability of passivating gel layers formed on International Simple Glass, *J. Nucl. Mater.* 524 (2019) 21–38.
- [33] D. Strachan, Glass dissolution as a function of pH and its implications for understanding mechanisms and future experiments, *Geochem. Cosmochim. Acta* 219 (2017) 111–123.
- [34] J.D. Vienna, J.J. Neeway, J.V. Ryan, et al., Impacts of glass composition, pH, and temperature on glass forward dissolution rate, *npj Mater. Degrad.* 2 (1) (2018).
- [35] S. Gin, P. Jollivet, M. Fournier, et al., The fate of silicon during glass corrosion under alkaline conditions: a mechanistic and kinetic study with the International Simple Glass, *Geochem. Cosmochim. Acta* 151 (2015) 68–85.
- [36] M.I. Ojovan, A. Pankov, W.E. Lee, The ion exchange phase in corrosion of nuclear waste glasses, *J. Nucl. Mater.* 358 (1) (2006) 57–68.
- [37] O.M. Farid, M.I. Ojovan, R.O. Abdel Rahman, Evolution of cations speciation during the initial leaching stage of alkali-borosilicate-glasses, *MRS Advances* 5 (3–4) (2020) 185–193.
- [38] T. Chave, P. Frugier, A. Ayral, et al., Solid state diffusion during nuclear glass residual alteration in solution, *J. Nucl. Mater.* 362 (2–3) (2007) 466–473.
- [39] P. Zapol, H. He, K.D. Kwon, et al., First-principles study of hydrolysis reaction barriers in a sodium borosilicate glass, *Int. J. Appl. Glass Sci.* 4 (4) (2013) 395–407.
- [40] K. Qin, B. Zhang, Z. Jin, et al., Influence of radiation on borosilicate glass leaching behaviors, *npj Mater. Degrad.* 8 (1) (2024) 9.
- [41] S. Zhu, Z. Jin, B. Zhang, et al., Electron irradiation-induced elements depletion and macroscopic changes in ternary borosilicate glasses: mechanism and composition dependence, *Appl. Surf. Sci.* (2024) 654.
- [42] M. Collin, S. Gin, P. Jollivet, et al., ToF-SIMS depth profiling of altered glass, *npj Mater. Degrad.* 3 (1) (2019).
- [43] L.T. Chen, X.T. Ren, Y.N. Mao, et al., Radiation effects on structure and mechanical properties of borosilicate glasses, *J. Nucl. Mater.* 552 (2021) 153025.
- [44] L.T. Chen, J.J. Mao, Y.N. Mao, et al., Changes in macroscopic and microscopic properties of borosilicate glass irradiated with ions, *Radiat. Eff. Defect Solid* 176 (9–10) (2021) 789–803.
- [45] J.P. Wan, J.S. Cheng, P. Lu, The coordination state of B and Al of borosilicate glass by IR spectra, *J. Wuhan Univ. Technol.-Materials Sci. Ed.* 23 (3) (2008) 419–421.
- [46] M.R. Filgueiras, G. LA Torre, L.L. Hench, Solution effects on the surface reactions of a bioactive glass, *J. Biomed. Mater. Res.* 27 (4) (1993) 445–453.
- [47] E.I. Kamitsos, M. Karakassides, G.D. Chrysikos, Vibrational spectra of magnesium-sodium-borate glasses. 2. Raman and mid-infrared investigation of the network structure, *JPhCh* 91 (5) (1987) 1073–1079.
- [48] X. Liu, M.N. Rahaman, D.E. Day, Conversion of melt-derived microfibrillar borate (13-93B3) and silicate (45S5) bioactive glass in a simulated body fluid, *JMSMM* 24 (2013) 583–595.
- [49] A. Tenney, J. Wong, Vibrational spectra of vapor-deposited binary borosilicate glasses, *J. Chem. Phys.* 56 (11) (1972) 5516–5523.
- [50] X. Zhang, J. Mao, B. Wang, et al., Evolution of mechanical properties and microstructure in borosilicate glasses with composition, *Eur. J. Glasses Sci. Technol. B Phys. Chem. Glasses* 62 (2021) 19–24.
- [51] X. Zhang, F. Yang, S. Zhu, et al., Influence of ion radiation on leaching behavior of borosilicate glass, *J. Non-Cryst. Solids* 602 (2023) 122091.
- [52] P. Gupta, A.C. Dillon, A.S. Bracker, et al., F₁s studies of H₂O and D₂O decomposition on porous silicon surfaces, *Surf. Sci.* 245 (3) (1991) 360–372.
- [53] T. Mueller, S. Schwertheim, W.R. Fahrner, Crystalline silicon surface passivation by high-frequency plasma-enhanced chemical-vapor-deposited nanocomposite silicon suboxides for solar cell applications, *JAP* 107 (1) (2010).
- [54] S. Kim, J. Park, P.D. Phong, et al., Improving the efficiency of rear emitter silicon solar cell using an optimized n-type silicon oxide front surface field layer, *Sci. Rep.* 8 (1) (2018) 10657.
- [55] C. LE Losq, D.R. Neuville, W. Chen, et al., Percolation channels: a universal idea to describe the atomic structure and dynamics of glasses and melts, *Sci. Rep.* 7 (1) (2017) 16490.
- [56] H. Arena, R. Podor, H.P. Brau, et al., Characterization of the boron profile and coordination in altered glass layers by EEL spectroscopy, *Micron* 141 (2021) 102983.
- [57] S. Gin, J.-M. Delaye, F. Angeli, et al., Aqueous alteration of silicate glass: state of knowledge and perspectives, *npj Mater. Degrad.* 5 (1) (2021).
- [58] K. Qin, Study on the Leaching Mechanism of Simple Borosilicate Glasses [D], Lanzhou University, Lanzhou, 2024.
- [59] A.H. Mir, High Heat Generating Wastes Research Forum 2024 [Z], 2024. Cambridge.